Physics of raindrops — Solution

X24 (2024) — English translation

A1 — 1.00 pts

Find the change in the free energy of water vapor if a drop of radius r is formed from it. Express your answer in terms of r, σ, φ, R, T, ρ_L, μ.

Due to the surface tension energy, the free energy will increase by

$$\Delta G_{surf} = \sigma A = 4\pi\sigma r^2.$$

On the other hand, for transition pair in liquid state it free energy decrease on

$$\Delta G_v = \nu RT \ln\varphi,$$

where the amount of substance in the droplet is

$$\nu = \frac{4\pi\rho_L r^3}{3\mu}.$$

Here we have used the formula for changing the free energy of vapor and the fact that for saturated vapor the free energy is equal to the free energy of the liquid.

Answer:

$$\Delta G = 4\pi\sigma r^2 - \frac{4\pi\rho_L}{3\mu} r^3 RT\ln\varphi$$

A2 — 0.80 pts

Find the critical value of the droplet radius r_c at which ΔG is maximum, as well as the corresponding value ΔG_c. Express your answer using σ, φ, R, T, ρ_L, μ. Find the numerical value of r_c at $\varphi = 1.01$.

To find the maximum, we find the derivative

$$\frac{\partial\Delta G}{\partial r} = 8\pi\sigma r - \frac{4\pi\rho_L}{\mu} r^2 RT\ln\varphi = 0.$$

Hence

$$r_c = \frac{2\sigma\mu}{\rho_L RT\ln\varphi}.$$

Substitute in formula for ΔG, obtain

$$\Delta G_c = \frac{16\pi}{3}\frac{\sigma^3\mu^2}{\rho_L^2 R^2 T^2 \ln^2\varphi} = \frac{4\pi r_c^2\sigma}{3}.$$

Answer:

$$r_c = \frac{2\sigma\mu}{\rho_L RT\ln\varphi} = 1.15\cdot 10^{-7}\ \text{m},\quad \Delta G_c = \frac{16\pi}{3}\frac{\sigma^3\mu^2}{\rho_L^2 R^2 T^2\ln^2\varphi}.$$

A3 — 0.70 pts

Consider a drop of critical radius r_c. Determine the time τ during which the number of molecules in it will increase by g. Express your answer using r_c, g, p_s, m, k, T, φ. Assume that during the growth process the radius of the drop does not change; the evaporation of molecules from the drop can be neglected.

It is known that

$$dN = dtdS\frac{p_v}{\sqrt{2\pi mkT}}$$

molecule. Here p_v – pressure pair, m – mass molecule, T, the gas temperature, falls on the surface area dS in time dt.

$$dN = 4\pi r_c^2\frac{p_v}{\sqrt{2\pi mkT}} = 4\pi r_c^2\frac{p_s\varphi}{\sqrt{2\pi mkT}}$$

molecule. Here use relation $p_v = p_s\varphi$. Since change radius one can neglect, coefficient proportionality constant, so the sought time is

$$\tau = g\left(4\pi r_c^2\frac{p_s\varphi}{\sqrt{2\pi mkT}}\right)^{-1}.$$

hits the entire surface area of the drop in time dt

Answer:

$$\tau = \frac{g\sqrt{2\pi mkT}}{4\pi r_c^2 p_s\varphi}.$$

A4 0.60 pts

Find the number of drops J that are formed per unit time in a unit volume of supersaturated water vapor. Express your answer in terms of σ, φ, p_s, r_c, T, g.

By condition, in time τ all nuclei in the volume turn into drops, so

$$J = \frac{n_c}{\tau} = \frac{4\pi r_c^2 p_s\varphi}{g\sqrt{2\pi mkT}}n\exp\left(-\frac{16\pi}{3kT}\frac{\sigma^3\mu^2}{\rho_L^2R^2T^2\ln^2\varphi}\right).$$

Express concentration pair through pressure:

$$n = \frac{p_v}{kT} = \frac{p_s\varphi}{kT},$$

we get

Answer:

$$J = \frac{4\pi r_c^2}{\sqrt{2\pi mkT}}\frac{p_s^2\varphi^2}{kT}\frac{1}{g}\exp\left(-\frac{16\pi}{3kT}\frac{\sigma^3\mu^2}{\rho_L^2R^2T^2\ln^2\varphi}\right) = \frac{4\pi r_c^2}{\sqrt{2\pi mkT}}\frac{p_s^2\varphi^2}{kT}\frac{1}{g}\exp\left(-\frac{4\pi r_c^2\sigma}{3kT}\right).$$

A5 0.90 pts

From the results of the previous paragraph it follows that the rate of droplet formation very much depends on the steam supersaturation coefficient. Determine numerically the value of the vapor supersaturation coefficient φ, at which one drop per second is born at a temperature $T = 283$ K in 1 cm^3 air. Consider that $g = 100$. The remaining numerical data are given at the beginning of the problem.

Let's substitute the expression for r_c into the result of the previous paragraph:

$$J = \frac{4\pi p_s^2}{\sqrt{2\pi mkT}}\frac{4\sigma^2\mu^2}{gkT\rho_L^2R^2T^2}\frac{\varphi^2}{\ln^2\varphi}\exp\left(-\frac{16\pi}{3kT}\frac{\sigma^3\mu^2}{\rho_L^2R^2T^2\ln^2\varphi}\right) = J_0\frac{\varphi^2}{\ln^2\varphi}\exp\left(-\frac{A}{\ln^2\varphi}\right),$$

where

$$J_0 = \frac{16\pi p_s^2}{\sqrt{2\pi mkT}}\frac{\sigma^2\mu^2}{gkT\rho_L^2R^2T^2} = 2.37\cdot 10^{30}\ \mathrm{m^{-3}\cdot s^{-1}},$$

$$A = \frac{16\pi}{3kT}\frac{\sigma^3\mu^2}{\rho_L^2R^2T^2} = 106.$$

We needed obtain value $J = 10^6\ \mathrm{m^{-3}\cdot s^{-1}}$. Numerically find $\varphi \approx 3.86$. Lead to also table value J for near value φ, we see that growth is happening very quickly.

φ J, $\mathrm{m^{-3}\cdot s^{-1}}$ 3.5 $8.4\cdot 10^1$ 3.6 $1.61\cdot 10^2$ 3.7 $2.34\cdot 10^4$ 3.8 $2.79\cdot 10^5$ 3.9 $2.68\cdot 10^6$

Answer:

$$\varphi = 3.86$$

B1 0.80 pts

For saturated vapor in equilibrium with a liquid, express the derivative of pressure with respect to temperature dp_s/dT in terms of p_s, L, R, T, μ. Using the result obtained, find the relative change in the density of saturated water vapor $\Delta\rho_s/\rho_s$ with a small change in temperature ΔT. Express your answer using ΔT, T, L, μ, R. You can use the relationship between small changes in pressure, density and temperature of an ideal gas

$$\frac{\Delta p_s}{p_s} = \frac{\Delta\rho_s}{\rho_s} + \frac{\Delta T}{T}.$$

The dependence of saturated steam pressure on temperature is determined by the Clapeyron-Clausius equation (we assume that the volume of steam is much larger than the corresponding volume of water at the same temperature):

$$\frac{dp_s}{dT} = \frac{L\mu p}{RT^2}.$$

Use relation from the condition, find

$$\frac{\Delta\rho_s}{\rho_s} = \frac{\Delta p_s}{p_s} - \frac{\Delta T}{T} = \frac{\lambda\mu p}{RT^2}\Delta T - \frac{\Delta T}{T} = \frac{\Delta T}{T}\left(\frac{\mu L}{RT} - 1\right).$$

Answer:

$$\frac{\Delta\rho_s}{\rho_s} = \frac{\Delta T}{T}\left(\frac{\mu L}{RT} - 1\right).$$

B2 0.20 pts

Express dQ/dt in terms of dM/dt and L.

Since heat is released only due to condensation of water $dQ = LdM$

Answer:

$$\frac{dQ}{dt} = L\frac{dM}{dt}$$

B3 0.30 pts

Using the result of the previous paragraph and the heat equation, express the temperature difference between the drop and the atmosphere, $T_r - T$, in terms of dM/dt, as well as r, L, K.

From the heat equation

$$T_r - T = \frac{1}{4\pi rK}\frac{dQ}{dt} = \frac{L}{4\pi rK}\frac{dM}{dt}.$$

Answer:

$$T_r - T = \frac{L}{4\pi rK}\frac{dM}{dt}.$$

B4 0.30 pts

We assume that near the surface of the drop the density of water vapor is equal to the density of saturated vapor at the temperature of the drop. Assuming the temperature and density differences are small and using the results B1, B3, express the ratio $(\rho_r - \rho_s)/\rho_s$ (ρ_r is the vapor pressure near the surface of the drop) in terms of L, r, K, μ, R, T and dM/dt.

From the result B1

$$\frac{\rho_r - \rho_s}{\rho_s} = \frac{T_r - T}{T}\left(\frac{\mu L}{RT} - 1\right).$$

Substituting the expression for the temperature difference into it, we obtain

Answer:

$$\frac{\rho_r - \rho_s}{\rho_s} = \left(\frac{\mu L}{RT} - 1\right) \frac{L}{4\pi r K T} \frac{dM}{dt}.$$

B5 — 0.30 pts

Using the diffusion equation, express the ratio $(\rho_r - \rho_v)/\rho_s$ in terms of dM/dt, r, D, ρ_s.

From the diffusion equation

$$\rho_v - \rho_r = \frac{1}{4\pi r D} \frac{dM}{dt},$$

Answer:

$$\frac{\rho_r - \rho_v}{\rho_s} = -\frac{1}{4\pi r \rho_s D} \frac{dM}{dt}$$

B6 — 0.60 pts

By excluding the vapor density near the surface of the drop ρ_r from the answers in the two previous paragraphs, you obtain an expression for dM/dt. Express your answer in terms of φ, μ, R, T, D, p_s, L, K, r.

Subtracting the expressions from the last two points from each other, we get

$$\frac{\rho_v - \rho_s}{\rho_s} = \varphi - 1 = \frac{1}{4\pi r} \left(\left(\frac{\mu L}{RT} - 1 \right) \frac{L}{KT} + \frac{1}{\rho_s D} \right) \frac{dM}{dt}.$$

Also express the saturated vapor density through pressure:

$$\rho_s = \frac{\mu p_s}{RT},$$

we finally get

Answer:

$$\frac{dM}{dt} = \frac{4\pi r(\varphi - 1)}{\left(\frac{\mu L}{RT} - 1\right)\frac{L}{KT} + \frac{RT}{\mu p_s D}}$$

B7 — 0.50 pts

The rate of increase in the radius of the drop has the form

$$\frac{dr}{dt} = \frac{\xi}{r^k}.$$

Determine k and ξ, express answer through φ, ρ_L, μ, R, T, D, p_s, L, K.

The mass of the drop is related to the radius by the relation

$$M = \frac{4\pi}{3} \rho_L r^3,$$

therefore

$$\frac{dM}{dt} = 4\pi \rho_L r^2 \frac{dr}{dt},$$

and means

$$\frac{dr}{dt} = \frac{1}{4\pi \rho_L r^2} \frac{dM}{dt} = \frac{1}{r \rho_L} \frac{\varphi - 1}{\left(\frac{\mu L}{RT} - 1\right)\frac{L}{KT} + \frac{RT}{\mu p_s D}}$$

Answer:

$$k = 1, \quad \xi = \frac{\varphi - 1}{\left(\frac{\mu L}{RT} - 1\right)\frac{L}{KT} + \frac{RT}{\mu p_s D}} \frac{1}{\rho_L}.$$

B8 — 0.50 pts

Find the dependence of the radius of the drop on time. The initial radius of the drop is r_0. Express your answer in terms of r_0, ξ, t.

Let's integrate the equation

$$r\frac{dr}{dt} = \xi,$$

obtain

$$\frac{r^2}{2} - \frac{r_0^2}{2} = \xi t,$$

Answer:

$$r(t) = \sqrt{r_0^2 + 2\xi t}.$$

B9 — 0.50 pts

Let the initial radius of the drop be $r_0 = 0.7$ μm. Find the numerical value of the time during which it will grow to size $r_1 = 10$ μm with a supersaturation coefficient $\varphi = 1.1$. The remaining numerical values are given at the beginning of this part.

For the parameters from the condition

$$\xi = 9.04 \cdot 10^{-12}\ \mathrm{m}^2 \cdot \mathrm{s}^{-1},$$

then the time

Answer:

$$t = \frac{r_1^2 - r_0^2}{2\xi} = 5.50\ \mathrm{s}.$$